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Weed and Pest Management Report

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1.0 Introduction

The purpose of this report is to outline methods for the control of weed and pest animal species within the bush and riparian planting areas that are proposed for covenanting under the terms of this consent. Weed invasion is highest in the bush edges bordering paddocks, but overall it is low. This plan outlines the weed species within the proposed covenant areas and provides appropriate methods for their initial and ongoing control. It also outlines control methods for animal pests that are likely to be, or are possibly, present.

2.0 Weed Control

Wed presence is as described in the accompanying Bush Quality Assessment and Riparian Revegetation Reports. Weed species observed within the proposed covenants include kikuyu, pine, foxglove, gorse, Japanese honeysuckle, willow, ivy, and pampas.

2.1 Control Methodology

Physical control should be undertaken, where possible, for small infestations, but on a large scale chemical control would be more efficient. It is recommended that spray applicators have completed a recognised agrichemical course and that application strictly follows the direction supplied with the chemicals. It is recommended that the appropriate safety equipment, including overalls and a respirator, are used.

Any weeds subsequently found or which may establish can be controlled as part of a single stage of works. The initial control of any weeds should be by the methods recommended in the table below. Long term eradication is the goal so weeds must be absent or kept to very low levels (controllable with a back pack sprayer used two times a year as per likely consent conditions).

Some of the weeds listed may require more than one spraying to bring them under control. When controlling climbers, avoid broadcast spraying that will affect the native vegetation they are climbing through. If they are present in the canopy the stems should be cut and the lower segment of the vine sprayed.

Weed Species Present within Proposed Covenant

Species	Herbicide	Knapsack /15L	Brushgun /100L	Comments
Blackberry	**Triclopyr (e.g. Grazon)**	45 – 90ml	300 – 600ml	Spray in spring and summer.
Foxglove	**Tordon Brushkiller**	37.5ml + penetrant	250ml + penetrant	Pull out and destroy small seedlings before flowering.
Gorse	**Triclopyr (e.g. Grazon)**	45 – 90ml	300 – 600ml	Spray in spring or summer when the plant is actively growing for best results. Higher doses to be used from February to May. Alternatively clear with brush cutter and mulch/burn dead material.

Ivy	**Glyphosate (e.g. Roundup)**	150ml (1%) + 30ml penetrant	1L (1%) + 200ml penetrant	
Japanese honeysuckle	**Triclopyr (e.g. Grazon)**	45 – 90ml	300 – 600ml	Spray in spring or summer when the plant is actively growing for best results. Higher doses to be used from February to May.
Kikuyu grass	**Glyphosate (e.g. Roundup)**	300ml (2%)	2L (2%)	Repeat spraying will be needed.
Pampas	**Glyphosate (e.g. Roundup)**	300ml (2%)	2L (2%)	Ensure thorough wetting of central area. Ensure spray drift does not occur.
Willow	**Glyphosate (e.g. Roundup)**	300ml (2%)	2L (2%)	Cut down larger trees or hand pull small examples. Paint stumps with 25% Roundup mix. Control by injection can also occur. Each stem must be treated separately.

Local Weed Species with the Potential to Invade and Control Methods

Species	**Herbicide**	**Knapsack /15l**	**Brushgun /100l**	**Comments**
Bindweed	**Triclopyr (e.g. Grazon)**	45 – 90ml	300 – 600ml	Spray lower part of vine. Ensure spray does not contact native trees.
Blackberry	**Triclopyr (e.g. Grazon)**	45 – 90ml	300 – 600ml	Spray in spring and summer.
Chinese privet	**Triclopyr (e.g. Grazon)**	45 – 90ml	300 – 600ml	Cut down larger trees or hand pull small examples. If cut down, use 25% mix on cut stump to prevent regrowth.
Cotoneaster	**Glyphosate (e.g. Roundup)**	300ml (2%)	2L (2%)	Cut down larger trees or hand pull small examples. Paint stumps with 25% Roundup mix.
Ginger	**Triclopyr (e.g. Grazon)**	45 – 90ml	300 – 600ml	Spray in spring or summer when the plant is actively growing for best results. Higher doses to be used from February to May.
Hawthorn	**Triclopyr (e.g. Grazon)**	45 – 90ml	300 – 600ml	Cut down larger trees or hand pull small examples. If cut down, use 25% mix on cut stump to prevent regrowth.
Himalayan honeysuckle	**Triclopyr (e.g. Grazon)**	45 – 90ml	300 – 600ml	
Tradescantia	**Triclopyr (e.g. Grazon)**	90ml + 15ml penetrant	600ml + 100ml penetrant	Spray in spring and summer.
Tree privet	**Triclopyr (e.g. Grazon)**	45 – 90ml	300 – 600ml	Cut down larger trees or hand pull small examples. If cut down, use 25% mix on cut stump to prevent

				regrowth.
Woolly nightshade	**Glyphosate (e.g. Roundup)**	150ml (1%)	1L (1%)	Cut tree down or spray/hand pull small examples. If cut down, use 25% mix on cut stump to prevent regrowth.

In regard to woody weeds such as gorse and willow, broadcast spraying of the leaves of larger individuals, when they are located within or adjacent native trees, should not occur due to the risk of widespread damage beyond the target plant. Rather the plant should be cut down and then the stump painted with a concentrated herbicide mix. For larger trees, injection will be best. Any regenerating seedlings or smaller examples can be hand-pulled. In open areas where mature plants have been common, significant regeneration from the seedbank can be expected, and hand-pulling is effective at keeping these under control if it is undertaken regularly.

2.2 Exotic tree control

Where there are exotic trees proposed to be within the covenanted area, it is recommended that they be ringbarked or poisoned, with poisoning being the preferred option. If ring barking is to occur then a continuous 5cm wide cut through the bark and sap wood all the way around the tree is needed. While this can be very effective, it is labour intensive on a large tree and it also has the effect of weakening the tree. Once it dies and loses strength such trees are more prone to being blown over than trees that have been poisoned. For this reason, poisoning is the recommended approach here.

If using Roundup, or other glyphosate based herbicide, drill 2cm diameter holes at a downward angle through the bark and about 50mm into the trunk, spacing the holes about 200mm apart around the trunk. A petrol powered auger is ideal for this job and these can be hired from hire centres. After holes are complete fill each hole with concentrated glyphosate. A drench gun can be used for injecting the herbicide. Die back should be observable within about three weeks.

Because high concentrations of herbicide are being used, safety equipment including gloves, goggles, respirator, chemical resistant footwear and cotton overalls should be used at all times.

2.3 Recommended chemicals and application

Glyphosate (e.g. Roundup) is the generally recommended herbicide when good results with its use can be obtained due to low toxicity and low residual activity. However, it is less effective on plants such as blackberry, and can have a significant non-target effect when used on other species. Triclopyr (e.g Grazon) is very effective against blackberry and other broadleaf weeds and trees, and has the advantage that it will not kill grasses which would open up the land to the threat of further invasion. When applying herbicide to a cut stump this must be done straight after cutting otherwise the stump will seal off and the herbicide will not penetrate properly.

All spraying should be undertaken in calm conditions to ensure spray drift is minimised. Spraying should also occur after and dew has evaporated and when rain is not likely for at least six hours. A marker dye added to the spray can be useful and lessens the likelihood of missing some weeds or spraying others twice, thus wasting spray. It can also highlight any spray drift that may be occurring.

2.4 Ongoing maintenance

Weed control should occur at the times specified in the following table. These times are mainly at the start of or during the main growth periods when weeds and competing grasses create the most problems. More intensive weed control will be required in the early years to control any regrowth. Gorse regrowth in areas where it is not currently present must be sprayed when it is very small, or the risk of damaging the adjacent native trees, which will be susceptible to the same sprays, will be much greater. A nozzle designed to minimise spray drift should be used for this job. Ongoing control will be required to keep all weeds at very low levels. Ideally control should always occur before they have reached a stage where they are producing seed themselves.

Annual maintenance cycle for weed control (*)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Year 1				*						*		*
Year 2		*		*						*		*
Year 3		*									*	
Year 4			*								*	
Year 5			*								*	

The covenants require the control of pests and weeds in perpetuity, continuing as per Year 5 of the above table. However, the initial years are likely to require the greatest effort. Control will be much easier, and it will be easier to target just weed species, if it is undertaken regularly so that the weeds being controlled are always kept small and before they set seed themselves. Weeds are most likely to establish in areas where there is plenty of sunlight such as on the edges and in the more open areas. These areas should therefore be the main focus for monitoring and control. Checks for weeds should be made on a biannual basis by walking the edges and more open areas to assess the success of previous control and check for any re-establishment. However, some weeds can establish and grow in quite deep shade. Therefore at least an annual check through the entire vegetation should be made for these species. In the shade they do not spread rapidly so new infestations should be relatively easily controllable if found after a year. It is much easier to eradicate a weed infestation while it is small than if it is left to spread widely.

A record of weed control effort and chemical use should be kept as per the following table (example).

Location in Covenant Area	Date of Weed Control	Species targeted	Chemical used/ Concentration/ physical control?	Method of application

3.0 Pest Control

Control of the pests present will allow for the ongoing development of the area as a wildlife habitat. Pest control will have large benefits for all other native species including insect life along with native geckos and skinks. Pest numbers should be minimised to a level where their impact on native wildlife is significantly reduced. Control of species that may be present is outlined below.

3.1 Possums

Options for possum control include poisoning, shooting and trapping. Poisoning is the recommended control method on this property as it is lower maintenance and therefore more suited to the large areas involved.

For poisoning, the bait stations and poisons can be obtained from rural supply shops. At least 53 bait stations (one per ha) located throughout the areas to be covenanted will provide suitable control for possums that may be present here. Bait should be kept fresh, with the baiting strictly following the operational instructions on the pack. Bait lines should be established throughout the bush areas for ease of access. The bait stations should be set 50 to 100m apart, with them being denser where there are wider areas between the bait station lines.

The recommended bait is either brodifacoum based (e.g. Pestoff waxed possum bait) or encapsulated cyanide (e.g. Feratox). Bait should be kept fresh, with the baiting strictly following the operational instructions on the pack. Initially, it should be placed in all bait stations and then checked on a weekly basis. If the bulk of the bait is being taken in that time, then refilling the stations should occur more frequently.

Following the initial baiting, baiting should again occur in March/April and July/August each year. The bait stations should be checked a week after filling and all untaken bait removed. If left it will degrade and become less effective, leading to possums that become sick and survive, which will create an aversion to bait stations.

3.2 Mustelids and rats

Likewise with the possums, poisoning is the recommended control method for rats on this property due to the large areas involved. Baiting should be undertaken using the same methodology as outlined for possums baiting, but with specialised rodent baits in bait stations (e.g. Philproof). Stations should be placed 50 to 100m apart in a grid pattern covering the areas to be protected. 100 rodent bait stations are recommended.

Mustelids will be controlled in part by feeding on poisoned rats, but it is also recommended some DoC200 traps are installed for additional control of these pest animals. They should be set inside commercially available purpose built DoC 200 boxes that mustelids enjoy to explore. Some camouflage of the tunnel is helpful. Suitable bait such as egg, meat or pet food will improve kill rates. Using cat or dog food risks the trapping of wandering pets, but the small opening of custom made housings almost eliminates this risk.

At least 15 traps should be placed within the areas to be covenanted – one in the new rural residential lots to cover the riparian planting areas, two in Area CV, two in Area CS, one in Area CT, and the other nine in Area CU. Mustelids are more likely to explore corridors or edges of wildlife habitat, so preference should be given to these locations for trap placement. As mustelids will constantly reintroduce ongoing control will be needed, but the most important time is prior to and during spring when many native bird species are breeding.

3.3 Monitoring success of trapping/baiting

A record of pest control effort should be kept as per the following table (example).

Location in Covenant Area	Date of Pest Control	Species targeted	Trap catch rate/Bait uptake rate

3.4 Goats, deer and pigs

If any sign of these pests is found on the property, control will need to occur. Hunting will need to provide the bulk of the effort, and a contract hunter (with dogs) employed every two years to ensure that numbers are sufficiently reduced is recommended.

For pigs, a hunter with dogs should go through the area on sign of them, or at least on an annual basis. The dogs should have GPS collars fitted, so that at the end of the hunt, the track can be downloaded and superimposed on a plan of the property, thus showing both coverage, and location of any kills. Records of all pig kills should be kept. Local pig hunting clubs should have people able to undertake the control needed.

For purposes of Council monitoring; records of cull numbers will be obtained from the contract hunter, and diary notes kept each time an animal is taken by a recreational hunter. These records can then be supplied to Council on an annual basis (following a contract hunt in April) or at any time as requested.

As there is no 'season' for the recreational hunting of goats the following method is recommended:

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Year 1	*	*	*	#								*
Year 2												
Year 3	*	*	*	#								*
Year 4												
Year 5	*	*	*	#								*

* = Bulk of recreational hunting effort
= Contract hunter employed to ensure hunting effort is adequate

While a concentrated hunting effort will be made in the summer months (prior to a contractor being employed) it is possible that hunting will occur on an ad-hoc basis throughout the year. After Year 5, control should be continued every two years in perpetuity as per the table. Annual hunting success should be recorded as per the following table.

Pest animal	Hunt date	Number killed	Sex	Approx. weight	Location (e.g. GPS coordinate)
Goat					
Pig					
Deer					

Signs that pigs or goats are present include browse of understorey vegetation and ground disturbance from rooting pigs. Goats often leave an obvious smell and hoof prints. Faeces are also an obvious sign of the presence of these pests.

4.0 Kauri Dieback

Kauri occur on the subject site and are currently free from kauri dieback so care must be taken to ensure this remains the case. Kauri Dieback (*Phytophthora agathidicida*) is known in the area where this property is located, but has not been recorded on the property itself. Even so, Kauri Dieback procedures should be followed when undertaking weed and pest management within the bush area, to avoid potential spread, as it may be present in the soil even if there are no signs of infection on the vegetation.

Management efforts for Kauri Dieback are largely focussed on limiting the spread within diseased forests and protecting healthy kauri areas. It is essential to maintain stock proof fencing around the bush area, as wild ungulates are major spreaders of the disease. Signage should also be erected on the property to indicate that kauri are present, especially if weed and pest control is to be undertaken by external contractors. This control should avoid kauri trees where possible, for example don't locate traps on kauri. Tree roots and the soil around the tree roots should be avoided where possible, and trigene spray used on all footwear before entering the area.

Signs of Kauri Dieback would include yellowing of foliage, loss of leaves, canopy thinning, dead branches and lesions that bleed gum at the base of the trunk. If any of these signs are noted, contact the Kauri Dieback Hotline on 0800 NZ KAURI (69 52874).

For further information, refer to www.kauridieback.co.nz.

5.0 Conclusion

Control of weeds on this property is manageable and the weeds should be controllable to Council standards. The initial two to three years are when the largest investment in weed control will be needed. Ongoing monitoring in perpetuity will be required to ensure that weed species never re-establish significant colonies before they are found and eradicated.

Several pest animal species are likely to be present and the proposed methodology will continue to bring numbers of possums and mustelids down to acceptable levels. Hunting will be employed for the control of goats, pigs and deer should any signs been found. If undertaken as outlined, this should keep numbers of these pests to acceptable levels.

6.0 Limitations

This report has been prepared for the particular project described to us and its extent is limited to the scope of work agreed between the client and Cato Bolam Consultants Limited.

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